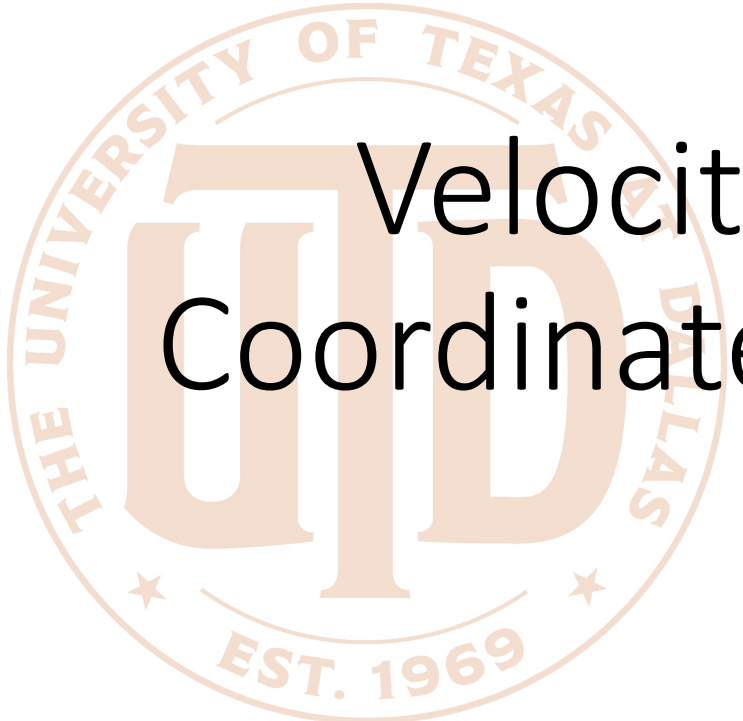


The logo of The University of Texas at Dallas is a circular seal. It features a large, stylized 'UTD' in the center. The words 'THE UNIVERSITY OF TEXAS' are written in a circle around the top, and 'AT DALLAS' is at the bottom. Below the 'UTD' is a banner that says 'EST. 1969'. There are two stars on either side of the banner.

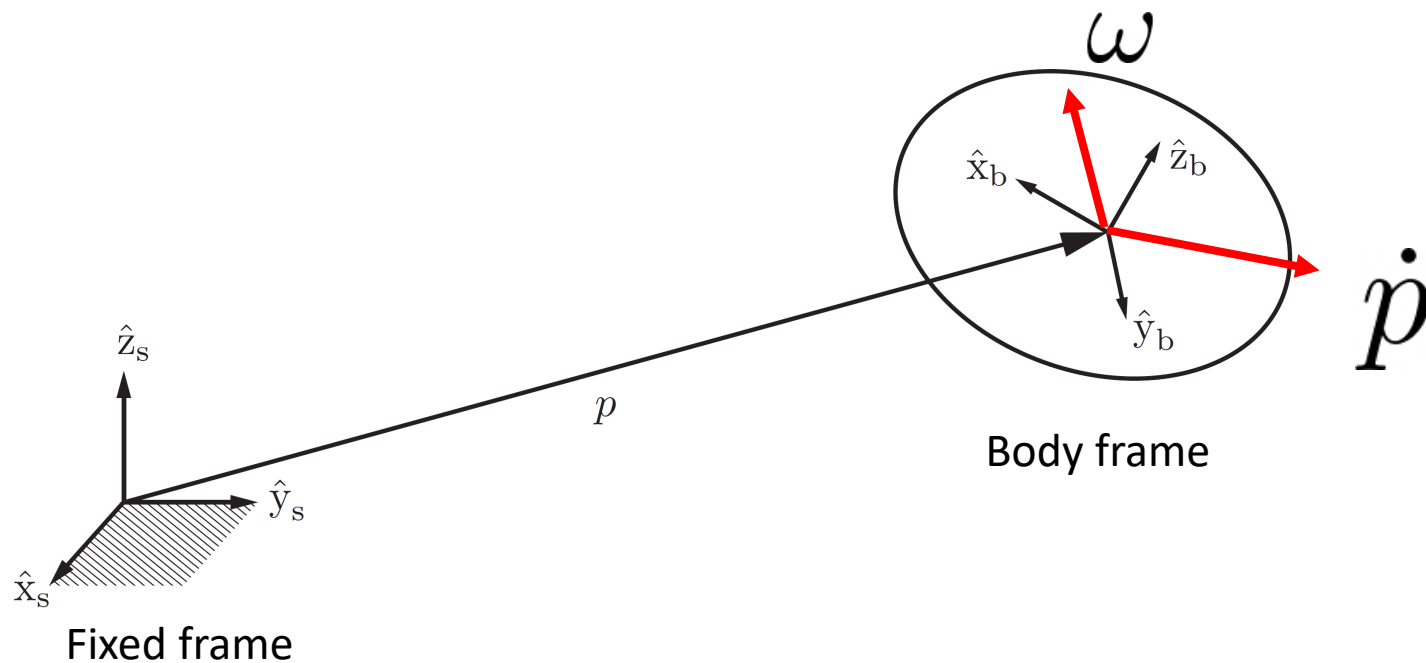
Velocity Kinematics: Exponential Coordinates of Rigid-Body Motions and Twists

CS 6341 Robotics

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The University of Texas at Dallas

Recall Angular Velocity and Linear Velocity



$$T_{sb}(t) = T(t) = \begin{bmatrix} R(t) & p(t) \\ 0 & 1 \end{bmatrix}$$

$$\dot{T} = \begin{bmatrix} \dot{R} & \dot{p} \\ 0 & 0 \end{bmatrix}$$

- Angular velocity

$$\omega = \hat{\omega} \theta$$

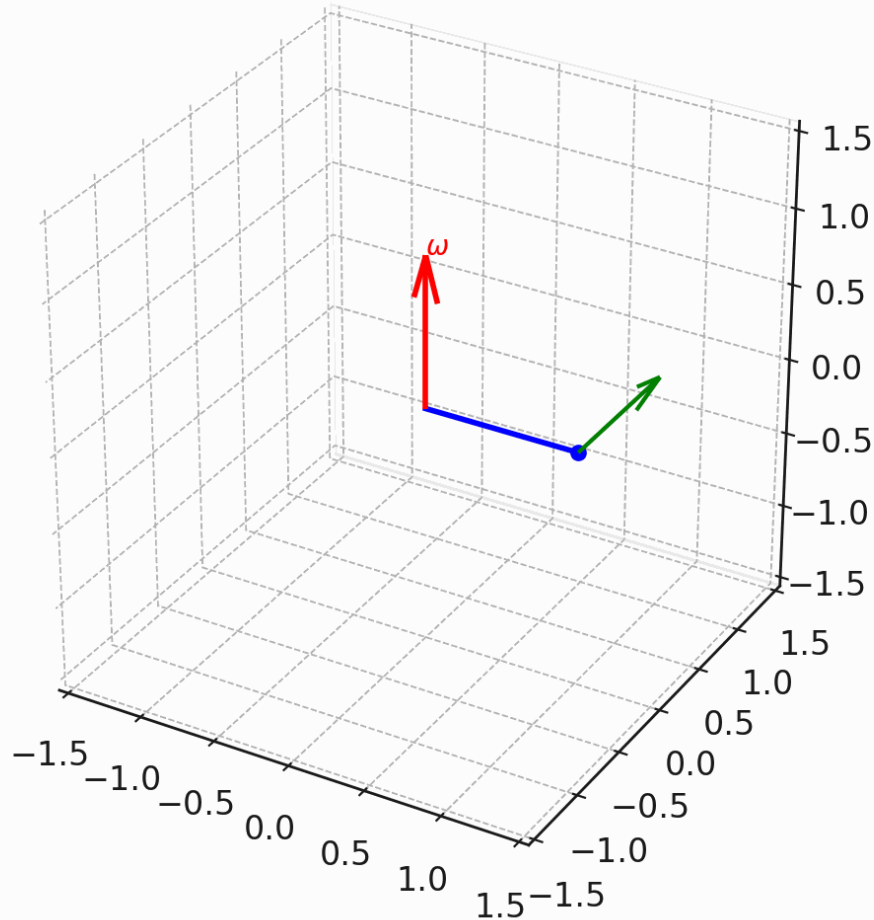
$$\dot{R}R^{-1} = [\omega_s]$$

$$R^{-1}\dot{R} = [\omega_b]$$

- Linear velocity $\dot{p}$

The linear velocity of the origin of {b} expressed in the fixed frame {s}

Angular Velocities



Generated by ChatGPT

- **Red arrow** $\rightarrow$ angular velocity vector ω
- **Blue line** $\rightarrow$ rotating body-fixed axis.
- **Green arrow** $\rightarrow$ instantaneous linear velocity of the blue endpoint ($\mathbf{v} = \omega \times \mathbf{r}$).

Tangential velocity
due to rotation

How to use angular velocity and linear velocity?

$$\dot{T} = \begin{bmatrix} \dot{R} & \dot{p} \\ 0 & 0 \end{bmatrix}$$

- How about this?

$$T(t + \Delta t) \approx T(t) + \dot{T}(t)\Delta t$$

- Adding $\dot{R}$ directly like this **breaks orthogonality** — the result won't generally be a valid rotation matrix.
- How to compute the transformation after certain time?

Exponential Coordinates of Rigid-Body Motions

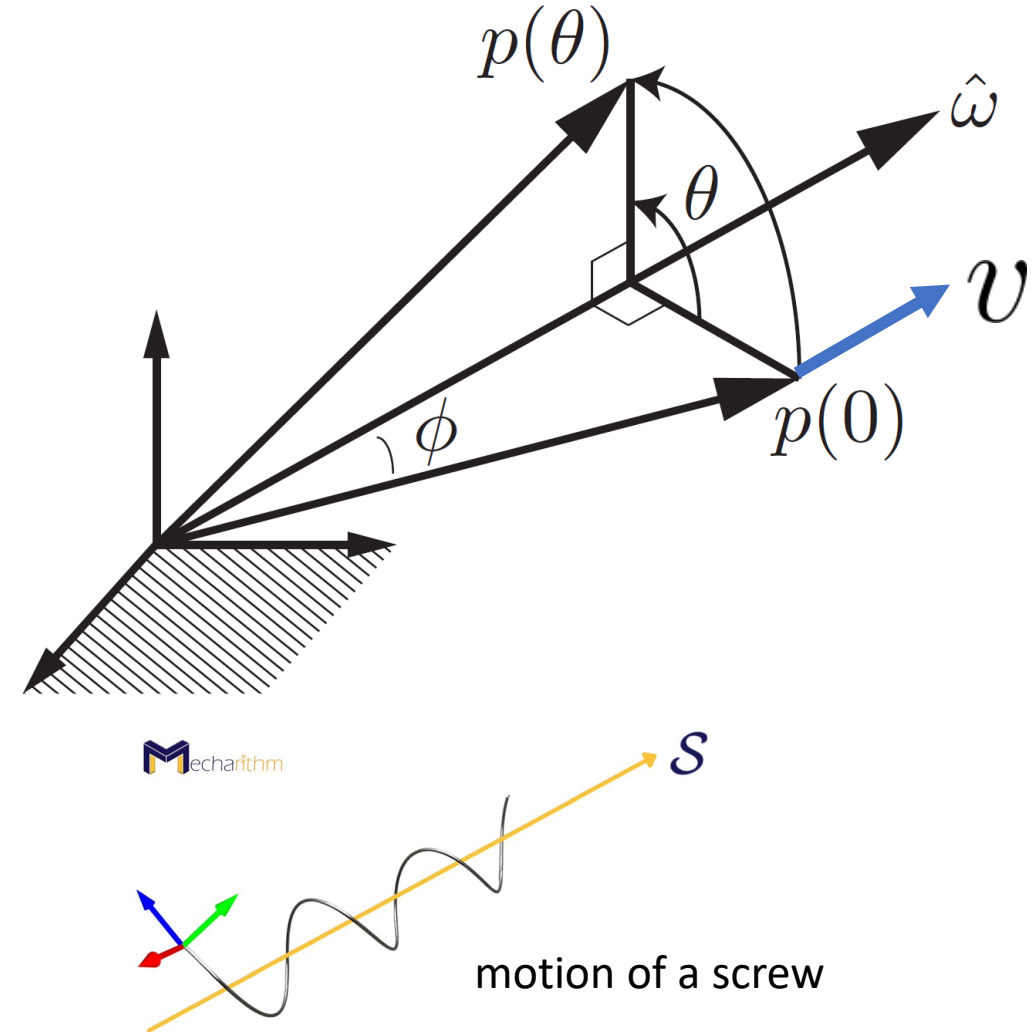
- $p(0)$ is rotated to $p(\theta)$
 - At a constant rate of 1 rad/s
- $p(t)$: path traced by the tip of vector

Velocity of the tip vector

$$\dot{p}(t) = v + \hat{\omega} \times p(t)$$

An additional
linear velocity

Tangential velocity
due to rotation



Exponential Coordinates of Rigid-Body Motions

- Linear Differential Equations $\dot{p}(t) = v + \hat{\omega} \times p(t)$
- Can we solve this equation to compute $p(t)$?
 - After time t , where is the vector?
- A scalar linear differential equation $\dot{x}(t) = ax(t)$ $x(t) \in \mathbb{R}, a \in \mathbb{R}$

Initial condition $x(0) = x_0$ Solution $x(t) = e^{at} x_0$

$$e^{at} = 1 + at + \frac{(at)^2}{2!} + \frac{(at)^3}{3!} + \dots$$

Exponential Coordinates of Rigid-Body Motions

- Vector linear differential equation

$$\dot{x}(t) = Ax(t) \quad x(t) \in \mathbb{R}^n, \quad A \in \mathbb{R}^{n \times n}$$

Initial condition $x(0) = x_0$ Solution $x(t) = e^{At}x_0$

matrix exponential
$$e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \dots$$

If A is constant and finite, this series converges to a finite limit

Exponential Coordinates of Rigid-Body Motions

- Linear Differential Equations $\dot{p}(t) = v + \hat{\omega} \times p(t)$
- Can we solve this equation to compute $p(t)$?
- Let's convert it to this form $\dot{x}(t) = Ax(t)$

$$\dot{\tilde{p}}(t) = \begin{bmatrix} [\hat{\omega}] & v \\ 0 & 0 \end{bmatrix} \tilde{p}(t) \quad \tilde{p}(t) = \begin{bmatrix} p(t) \\ 1 \end{bmatrix} \quad [\mathcal{S}] = \begin{bmatrix} [\hat{\omega}] & v \\ 0 & 0 \end{bmatrix} \in se(3)$$

Homogenous coordinate

Exponential Coordinates of Rigid-Body Motions

- Linear Differential Equations $\dot{p}(t) = v + \hat{\omega} \times p(t)$

$$\dot{\tilde{p}}(t) = [\mathcal{S}] \tilde{p}(t) \quad [\mathcal{S}] = \begin{bmatrix} [\hat{\omega}] & v \\ 0 & 0 \end{bmatrix}$$

- Solution $\tilde{p}(t) = e^{[\mathcal{S}]t} \tilde{p}(0)$

Homogenous transformation $T(t) = e^{[\mathcal{S}]t}$

matrix exponential $e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \dots$

What is this??

How to compute a 4 x 4 matrix from this?

Exponential Coordinates of Rigid-Body Motions

- Let's do some computation $T(t) = e^{[S]t}$
- We can change t to θ

$$[S] = \begin{bmatrix} [\hat{\omega}] & v \\ 0 & 0 \end{bmatrix}$$

$$\begin{aligned} T(\theta) &= e^{[S]\theta} = I + [S]\theta + [S]^2 \frac{\theta^2}{2!} + [S]^3 \frac{\theta^3}{3!} + \dots \\ &= I + \begin{bmatrix} [\hat{\omega}] & v \\ 0 & 0 \end{bmatrix} \theta + \begin{bmatrix} [\hat{\omega}] & v \\ 0 & 0 \end{bmatrix}^2 \frac{\theta^2}{2!} + \begin{bmatrix} [\hat{\omega}] & v \\ 0 & 0 \end{bmatrix}^3 \frac{\theta^3}{3!} + \dots \\ &= I + \begin{bmatrix} [\hat{\omega}] & v \\ 0 & 0 \end{bmatrix} \theta + \begin{bmatrix} [\hat{\omega}]^2 & [\hat{\omega}]v \\ 0 & 0 \end{bmatrix} \frac{\theta^2}{2!} + \begin{bmatrix} [\hat{\omega}]^3 & [\hat{\omega}]^2 v \\ 0 & 0 \end{bmatrix} \frac{\theta^3}{3!} + \dots \end{aligned}$$

Exponential Coordinates of Rigid-Body Motions

$$T(\theta) = e^{[S]\theta} = \begin{bmatrix} R(\theta) & G(\theta)v \\ 0 & 1 \end{bmatrix}$$

$$R(\theta) = I + [\hat{\omega}]\theta + [\hat{\omega}]^2 \frac{\theta^2}{2!} + [\hat{\omega}]^3 \frac{\theta^3}{3!} + \dots$$

$$G(\theta) = I\theta + [\hat{\omega}] \frac{\theta^2}{2!} + [\hat{\omega}]^2 \frac{\theta^3}{3!} + \dots$$

We have $[\hat{\omega}]^3 = -[\hat{\omega}]$

Exponential Coordinates of Rigid-Body Motions

$$\begin{aligned} R(\theta) &= I + [\hat{\omega}]\theta + [\hat{\omega}]^2 \frac{\theta^2}{2!} + [\hat{\omega}]^3 \frac{\theta^3}{3!} + \dots \\ &= I + \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right) [\hat{\omega}] + \left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \dots \right) [\hat{\omega}]^2 \end{aligned}$$

$$\begin{aligned} \sin \theta &= \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \\ \cos \theta &= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots \end{aligned}$$

$$R(\theta) = e^{[\hat{\omega}]\theta}$$

$$R(\theta) = I + \sin \theta [\hat{\omega}] + (1 - \cos \theta) [\hat{\omega}]^2$$

Rodrigues' formula: exponential coordinates to rotation matrix

Exponential Coordinates of Rigid-Body Motions

$$\begin{aligned} G(\theta) &= I\theta + [\hat{\omega}] \frac{\theta^2}{2!} + [\hat{\omega}]^2 \frac{\theta^3}{3!} + \dots & [\hat{\omega}]^3 &= -[\hat{\omega}] \\ &= I\theta + \left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \dots \right) [\hat{\omega}] + \left(\frac{\theta^3}{3!} - \frac{\theta^5}{5!} + \frac{\theta^7}{7!} - \dots \right) [\hat{\omega}]^2 \\ &= I\theta + (1 - \cos \theta) [\hat{\omega}] + (\theta - \sin \theta) [\hat{\omega}]^2 \end{aligned}$$

Exponential Coordinates of Rigid-Body Motions

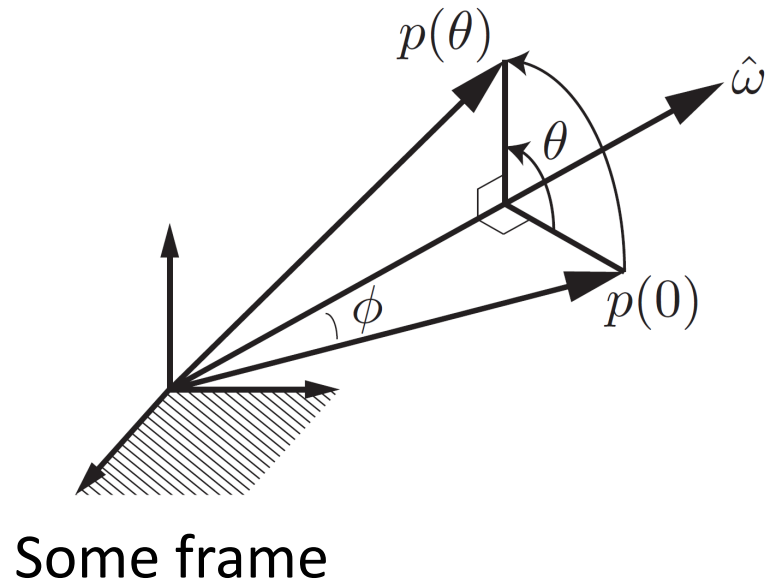
$$T(\theta) = e^{[S]\theta} = \begin{bmatrix} R(\theta) & G(\theta)v \\ 0 & 1 \end{bmatrix} \quad [S] = \begin{bmatrix} [\hat{\omega}] & v \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} I + \sin \theta [\hat{\omega}] + (1 - \cos \theta) [\hat{\omega}]^2 & (I\theta + (1 - \cos \theta) [\hat{\omega}] + (\theta - \sin \theta) [\hat{\omega}]^2)v \\ 0 & 1 \end{bmatrix}$$

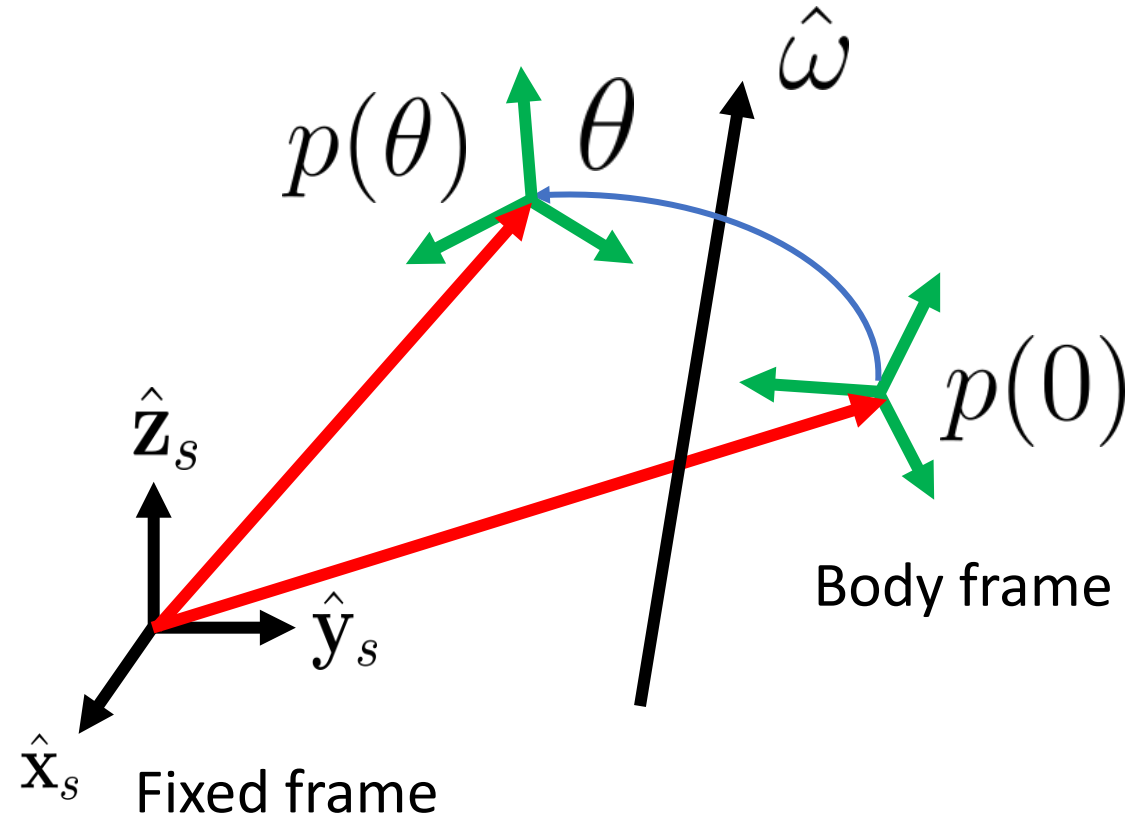
Conclusion: given unit angular velocity $\hat{\omega}$ and linear velocity v
 Use the above equation to compute the homogenous transformation after θ if $\|\omega\| = 1$

$$\text{If } \omega = 0 \text{ and } \|v\| = 1 \quad T(\theta) = e^{[S]\theta} = \begin{bmatrix} I & v\theta \\ 0 & 1 \end{bmatrix}$$

What if the rotation axis not going through the origin?



Case 1 (so far)



Case 2

What if the rotation axis not going through the origin?

- $p(0)$ is rotated to $p(\theta)$
 - At a constant rate of 1 rad/s
- $p(t)$: path traced by the tip of vector

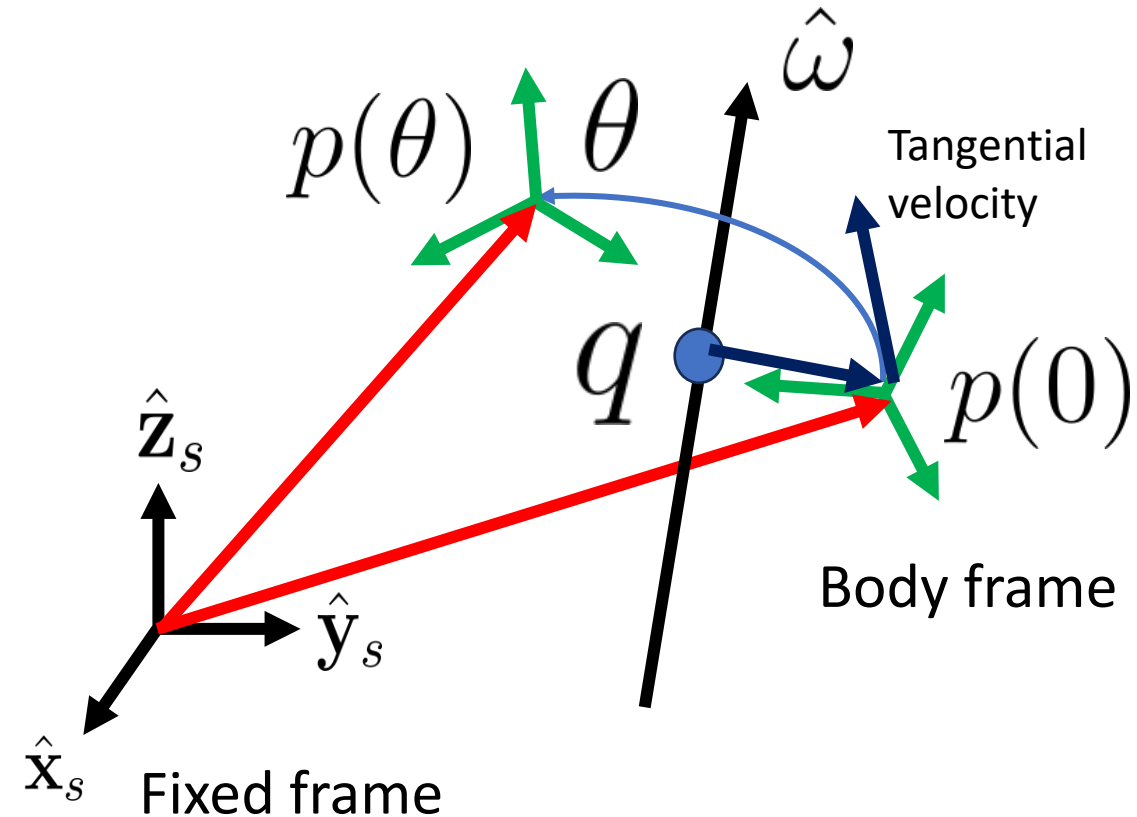
Velocity of the tip vector

$$\dot{p}(t) = v + \hat{\omega} \times (p(t) - q)$$

An additional
linear velocity

Tangential velocity
due to rotation

$$\dot{p}(t) = -\hat{\omega} \times q + v + \hat{\omega} \times p(t)$$



What if the rotation axis not going through the origin?

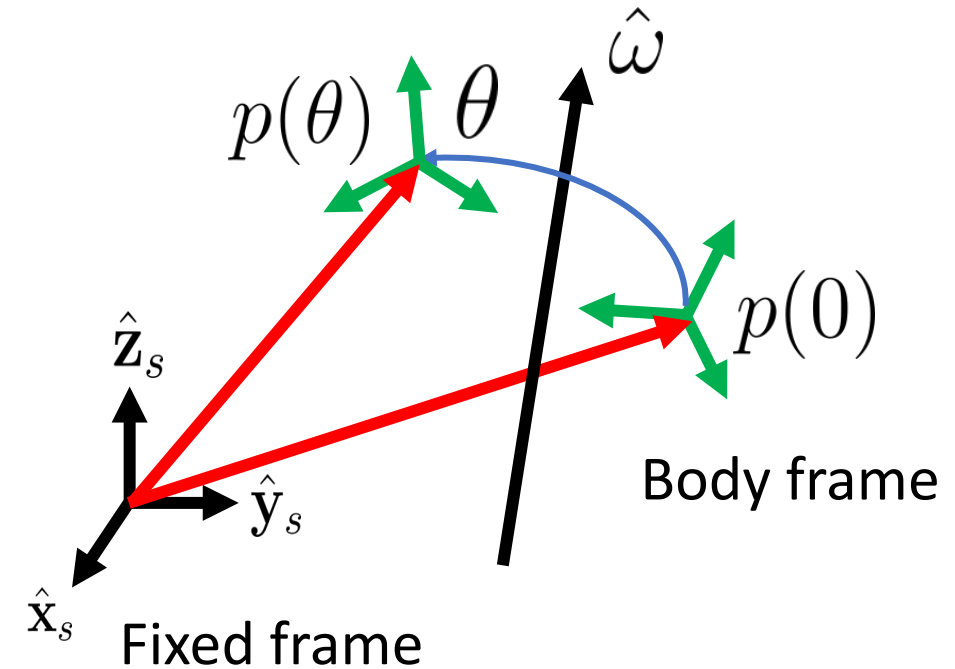
$$\dot{p}(t) = -\hat{\omega} \times q + v + \hat{\omega} \times p(t)$$

- Angular velocity $\hat{\omega}$
- Linear velocity

$$v_1 = -\hat{\omega} \times q + v$$

q can be any point on the rotation axis

$$T(\theta) = e^{[S]\theta} \quad [S] = \begin{bmatrix} [\hat{\omega}] & v_1 \\ 0 & 0 \end{bmatrix}$$



Twist

- Let's combine angular velocity and linear velocity into a 6D vector called twist

$$\mathcal{V} = \begin{bmatrix} \omega \\ v \end{bmatrix}$$

- Twist can be defined in fixed frame or body frame

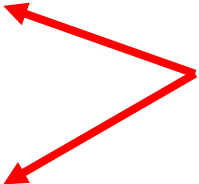
Spatial twist $\mathcal{V}_s = \begin{bmatrix} \omega_s \\ v_s \end{bmatrix} \in \mathbb{R}^6$

Body twist $\mathcal{V}_b = \begin{bmatrix} \omega_b \\ v_b \end{bmatrix} \in \mathbb{R}^6$

Relationship between Spatial Twist and Body Twist

- For angular velocity $\omega_s = R\omega_b$ $T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}$

- For linear velocity

$$\begin{aligned} v_s &= -\omega_s \times q_s + v_s^0 \\ v_b &= -\omega_b \times q_b + v_b^0 \end{aligned}$$


An additional linear velocity

$$q_s = Rq_b + p \quad v_s^0 = Rv_b^0$$

Relationship between Spatial Twist and Body Twist

- Some derivation

$$\begin{aligned}v_s &= -\omega_s \times q_s + v_s^0 \\&= -[R\omega_b](Rq_b + p) + Rv_b^0 \\&= -R[\omega_b]R^T(Rq_b + p) + Rv_b^0 & R[\omega]R^T = [R\omega] \\&= R(-[\omega_b]q_b + v_b^0) - R[\omega_b]R^T p \\&= Rv_b - [R\omega_b]p \\&= [p]R\omega_b + Rv_b & [\omega]p = -[p]\omega\end{aligned}$$

Relationship between Spatial Twist and Body Twist

$$\begin{bmatrix} \omega_s \\ v_s \end{bmatrix} = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \begin{bmatrix} \omega_b \\ v_b \end{bmatrix}$$



$$[\text{Ad}_T] = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \in \mathbb{R}^{6 \times 6}$$

The adjoint representation of $T = (R, p) \in SE(3)$ $\mathcal{V}_s = [\text{Ad}_{T_{sb}}] \mathcal{V}_b$

Relationship between Spatial Twist and Body Twist

$$T^{-1} = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix}$$

$$R[\omega]R^T = [R\omega]$$

$$\mathcal{V}_s = \begin{bmatrix} \omega_s \\ v_s \end{bmatrix} = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \begin{bmatrix} \omega_b \\ v_b \end{bmatrix} = [\text{Ad}_{T_{sb}}]\mathcal{V}_b$$

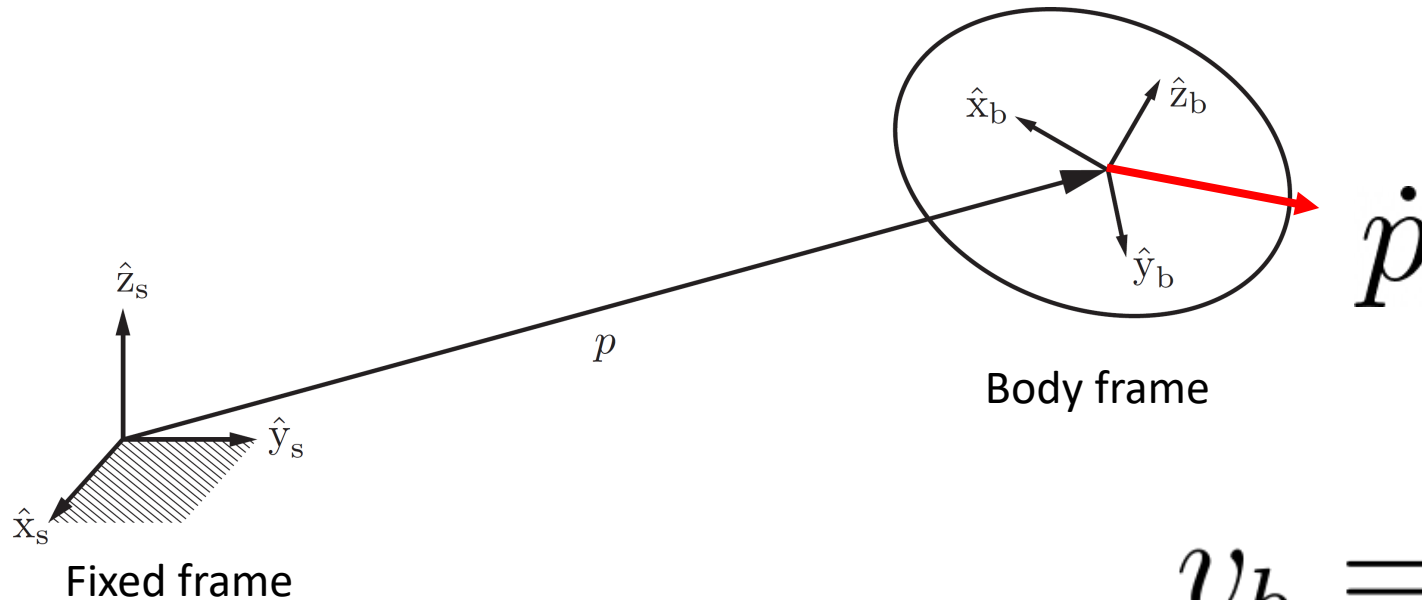
$$\mathcal{V}_b = \begin{bmatrix} \omega_b \\ v_b \end{bmatrix} = \begin{bmatrix} R^T & 0 \\ -R^T[p] & R^T \end{bmatrix} \begin{bmatrix} \omega_s \\ v_s \end{bmatrix} = [\text{Ad}_{T_{bs}}]\mathcal{V}_s$$

In general

$$\mathcal{V}_c = [\text{Ad}_{T_{cd}}]\mathcal{V}_d, \quad \mathcal{V}_d = [\text{Ad}_{T_{dc}}]\mathcal{V}_c$$

Relationship between Spatial Twist and Body Twist

- What is the relationship between v_s , v_b and $\dot{p}$? $\dot{T} = \begin{bmatrix} \dot{R} & \dot{p} \\ 0 & 0 \end{bmatrix}$



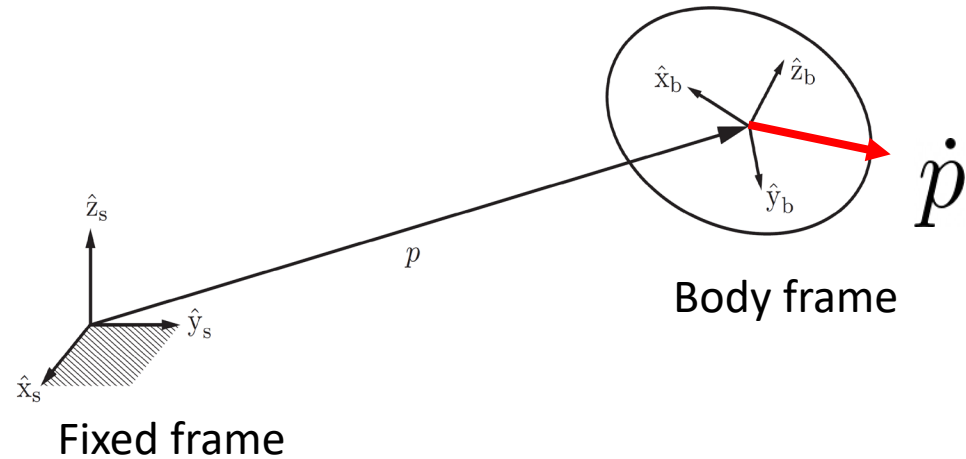
- Linear velocity $\dot{p}$

The linear velocity of the origin of {b} expressed in the fixed frame {s}

$$v_b = R_{bs}\dot{p} = R^T \dot{p}$$

Relationship between Spatial Twist and Body Twist

- What is the relationship between v_s , v_b and $\dot{p}$? $\dot{T} = \begin{bmatrix} \dot{R} & \dot{p} \\ 0 & 0 \end{bmatrix}$



- Linear velocity $\dot{p}$

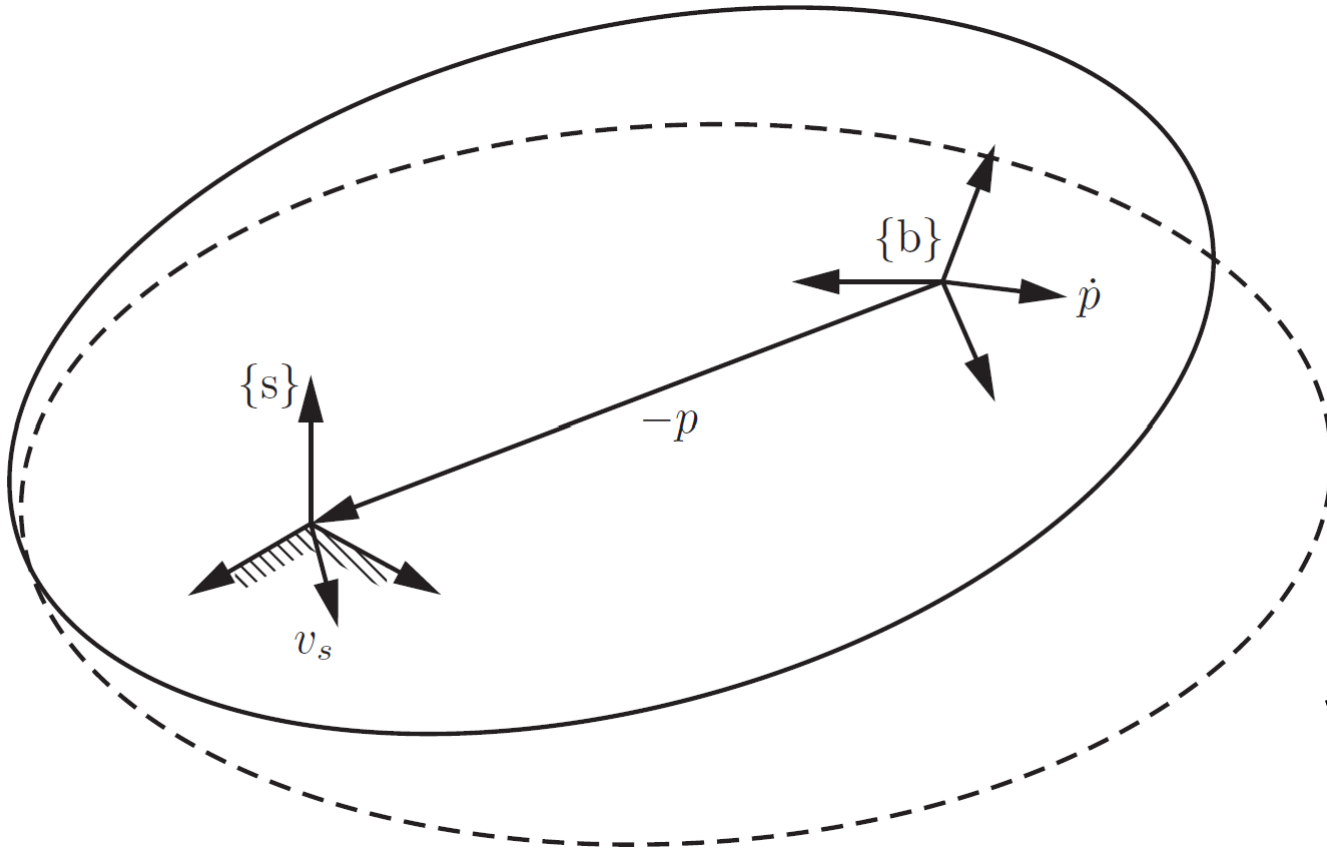
The linear velocity of the origin of {b} expressed in the fixed frame {s}

$$v_b = R_{bs}\dot{p} = R^T\dot{p}$$

$$v_s = [p]R\omega_b + Rv_b = [p]\omega_s + RR^T\dot{p}$$

$$v_s = \dot{p} + \omega_s \times (-p) \quad [\omega]p = -[p]\omega$$

Relationship between Spatial Twist and Body Twist



$$v_s = \dot{p} + \omega_s \times (-p)$$

v_s is the instantaneous velocity of the point on this body currently at the fixed-frame origin, expressed in the fixed frame

Summary

- Exponential Coordinates of Rigid-Body Motions
- Twists
 - Spatial twists
 - Body twists

Further Reading

- Chapter 3 in Kevin M. Lynch and Frank C. Park. Modern Robotics: Mechanics, Planning, and Control. 1st Edition, 2017